Who wins NBA MVP?

Predicting vote share with walk-forward models, compared with a Stanford CS229 reference

Machine learning project | NBA MVP prediction

Introduction

- The MVP is not a pure stat leaderboard. Around 100 media members vote, and each player ends up with a share of the possible points.
- Voters weigh scoring, efficiency and impact, but also team success and the size of a player's role.
- The question is whether those patterns are consistent enough to learn. If they are, past seasons should help us predict later ones.
- Using data from 1980–2025, this project evaluates predictions on held-out seasons from 1990–2025.

Problem statement

The MVP is decided by media vote, so the outcome mixes individual stats with team success and narrative. This project builds a regression model on historical seasons, ranks recorded vote recipients by predicted vote share, and picks the top-ranked player as the predicted MVP.

- Input: one row per MVP vote recipient per season (box + advanced stats + team context)
- Output: predicted vote share per player, then ranked inside each season
- Prediction: the player with the highest predicted share is the MVP pick

Database

- Source: public master_table.csv (DavidYoo912/nba_mvp), scraped from Basketball-Reference. Covers every player who received MVP votes from 1980–2025.
- The notebook downloads the CSV directly (no local database). After cleaning: 724 player-seasons, 45 original columns.
- Each row is one player in one season: age, games, minutes, points, rebounds, assists, steals, blocks, shooting percentages,
- PER, true shooting, win shares, BPM, VORP, usage rate, plus team wins, win %, and playoff seed.
- Target is Share — the fraction of possible MVP voting points. Only vote recipients are in the table.

Cleaning rules

No season filter (source starts in 1980). Drop missing Share or Player and 1 duplicate row (725 → 724); check for ties at the maximum share.

No extra games filter beyond what the source already contains.

Leakage columns (Rank, First, Pts Won, Pts Max, Share) are blocked from features.

Missing values are filled with the median inside each model pipeline.

Solution: how the model works

1. Features

32 numeric predictors: Age, G, MP, PTS, TRB, AST, STL, BLK, FG%, 3P%, FT%, WS, WS/48, W, W/L%, seed, PER, TS%, 3PAr, FTr, rebound/assist/steal/block %, TOV%, USG%, OBPM, DBPM, BPM, VORP.

2. Feature screening

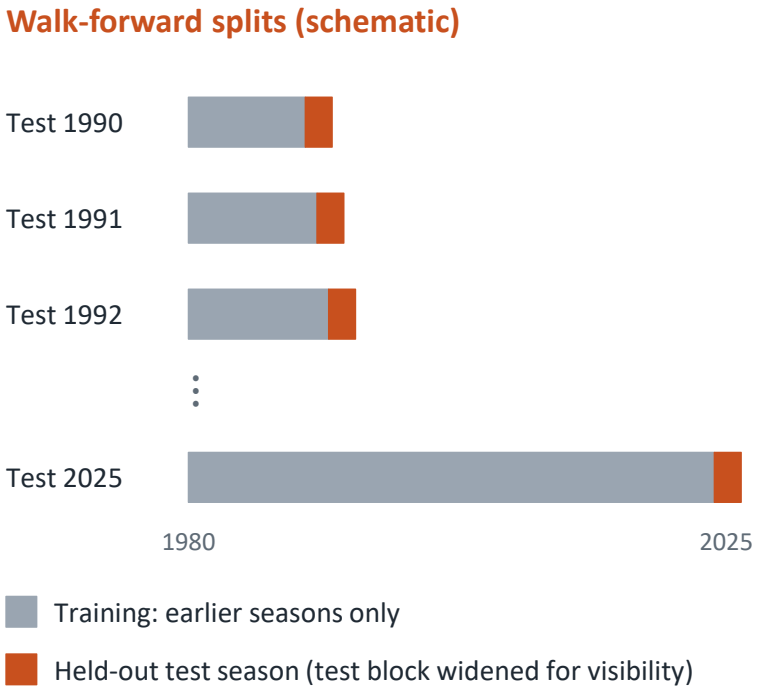
Identifiers, vote columns (Share, Rank, First, Pts Won/Max) and any column containing 'share'/'vote'/'mvp' are blocked. Only non-voting predictor columns are used; season-level statistics are available after the season.

3. Target

Continuous regression on Share (0–1). Predictions clipped to [0,1], ranked within the season; highest predicted share = predicted MVP.

Solution: models and testing

- Walk-forward: for each test season, train only on earlier seasons, then predict that season.
- Test seasons 1990–2025 (36 in total), after an initial 10-season training history.
- Models: Ridge (α = 10), Random Forest (400 trees), HistGradientBoosting (250 iterations).
- Baselines: prior-season vote share and Win Shares-only Ridge.
- MVP voting outcomes (Share, Rank, First, Pts Won/Max) are excluded; imputation and scaling are fit on training seasons only.
- Hyperparameters are fixed in the notebook: no tuning loop and no separate validation split.



How results are measured

Winner accuracy

Primary metric. Is the model's highest predicted share the actual MVP? Counted over 36 seasons.

Top-3 hit rate

Is the actual MVP among the model's three highest predicted shares in that season?

Spearman correlation

Rank agreement between predicted and actual vote share within a season, averaged over 36 seasons.

MAE / RMSE

How far predicted vote shares are from actual Share across all player-seasons. RMSE punishes big misses more.

Wilson 95% CI

Uncertainty band around winner accuracy. It is wide with only 36 seasons.

Paired season tests

Exact binomial test on the seasons where two models disagree.

These are definitions only. Results follow on slides 8–10, all on the same 36 held-out seasons.

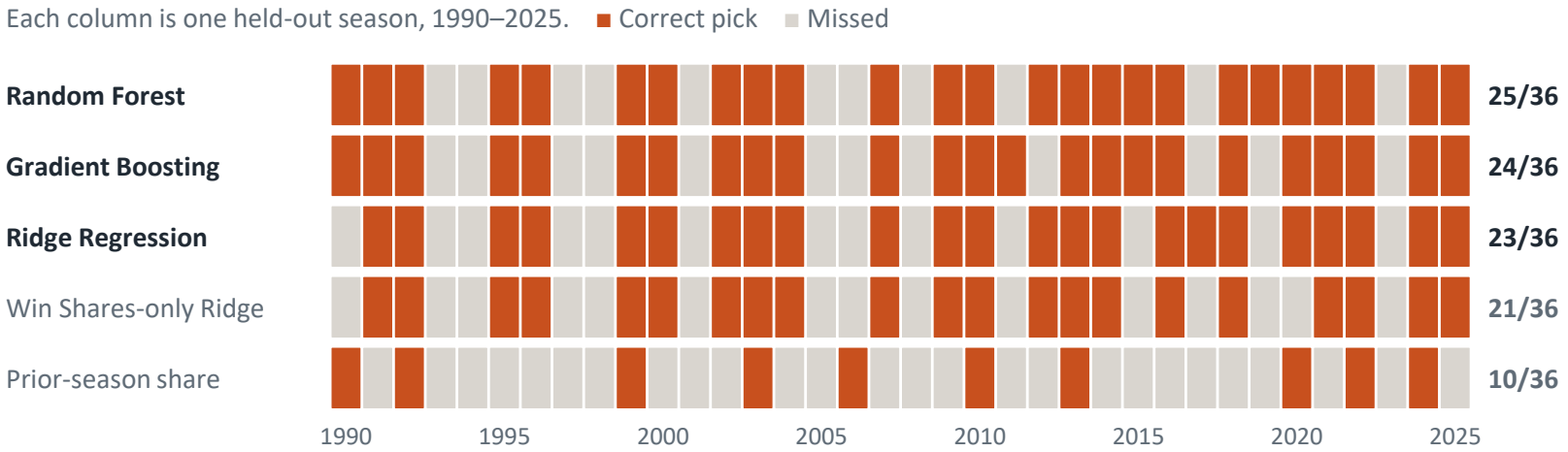
Results on 36 held-out seasons

Model	Winner accuracy	Top-3 hit rate	Spearman ρ	MAE	RMSE
Random Forest	**69.4% (25/36)**	88.9% (32/36)	0.73	0.120	0.176
Gradient Boosting	66.7% (24/36)	88.9% (32/36)	0.71	**0.109**	**0.172**
Ridge Regression	63.9% (23/36)	**94.4% (34/36)**	**0.78**	0.132	0.182
Baseline: Win Shares-only Ridge	58.3% (21/36)	77.8% (28/36)	0.64	0.156	0.218
Baseline: prior-season share	27.8% (10/36)	66.7% (24/36)	0.39	0.168	0.285

Best value in each column in orange. Spearman is the mean of per-season rank correlations. MAE and RMSE are on vote share (0–1); lower is better.

No single model wins every metric. Random Forest picks the MVP most often, Ridge ranks the field best (top-3 and Spearman), and Gradient Boosting has the smallest vote-share error.

Season by season: where models hit and miss



Consensus hits and misses

All three regression models were right in 21 seasons and wrong in 9, e.g. 1997–98 and 2023. Random Forest ranked Steve Nash 11th (2005) and 10th (2006).

Latest held-out season: 2025

Random Forest, predicted vs actual share: Gilgeous-Alexander 0.92 vs 0.91, Jokić 0.80 vs 0.79, Antetokounmpo 0.41 vs 0.47, Tatum 0.12 vs 0.31.

How strong is the evidence?

Winner accuracy with Wilson 95% interval

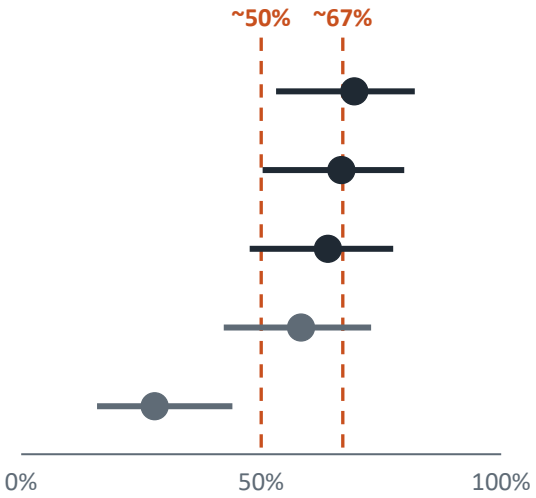
Random Forest
69.4% [53–82]

Gradient Boosting
66.7% [50–80]

Ridge Regression
63.9% [48–78]

WS-only Ridge
58.3% [42–73]

Prior-season share
27.8% [16–44]



Paired season tests (A vs B)

Comparison	Seasons A–B	One-sided p
RF vs prior-season	16–1	**0.0001**
GB vs prior-season	15–1	**0.0003**
Ridge vs prior-season	15–2	**0.0012**
RF vs WS-only Ridge	4–0	0.0625
GB vs WS-only Ridge	4–1	0.1875
Ridge vs WS-only Ridge	2–0	0.2500
GB vs RF	1–2	0.8750

RF = Random Forest, GB = Gradient Boosting, WS-only = Win Shares-only. A–B = seasons only A / only B was correct.

Clear: all three models beat the prior-season baseline ($p < 0.002$). **Not clear:** none is significantly better than Win Shares-only Ridge or than each other. Random Forest's 53–82% interval spans both Stanford figures (dashed lines).

Comparing with Stanford CS229

Study and setting	Candidates ranked	Test seasons	Winner accuracy
Stanford CS229 (as reported)	Top-3 finalists	4 seasons, fixed split	Naive Bayes ≈ 67% Logistic ≈ 50%
This project: full pool	All vote recipients (about 15 per season)	36 walk-forward seasons	**Random Forest 69.4%** **Gradient Boosting 66.7%** **Ridge 63.9%**
This project: finalist-only*	Actual top-3 vote getters	Same 36 seasons	**Random Forest 69.4%** **Gradient Boosting 66.7%** **Ridge 63.9%**

*Finalists are picked using the actual vote result, so this approximates Stanford's setup rather than replicating it. Restricting to finalists did not change top-1 accuracy for the three models.

How to read this: our accuracy is numerically close to Stanford's Naive Bayes figure, but the datasets, candidate sets and validation differ, so it is a reference, not a head-to-head. We do not claim superiority.

Conclusion

69.4%
Random Forest winner accuracy (25 of 36 seasons)

34 / 36
Seasons with the MVP in Ridge's top three (others: 32/36)

+41.7 pts
Random Forest over prior-season baseline ($p < 0.001$)

- Ranking predicted vote shares picks the MVP in roughly two of three held-out seasons, and the winner is almost always in the top three.
- All three models clearly beat the prior-season baseline, but none is significantly better than Win Shares-only Ridge or than each other.
- Stanford figures are a reference, not a head-to-head: different task, seasons and candidate set.
- Limits: voter narratives, injuries and award fatigue are unmeasured; only vote recipients are modelled; hyperparameters are fixed with no validation split.
- Next: prior-year vote share as a feature, fuller team context, and a true prospective forecast for the live season.